

FINA2204 Tutorial 7: Option Trading Strategies

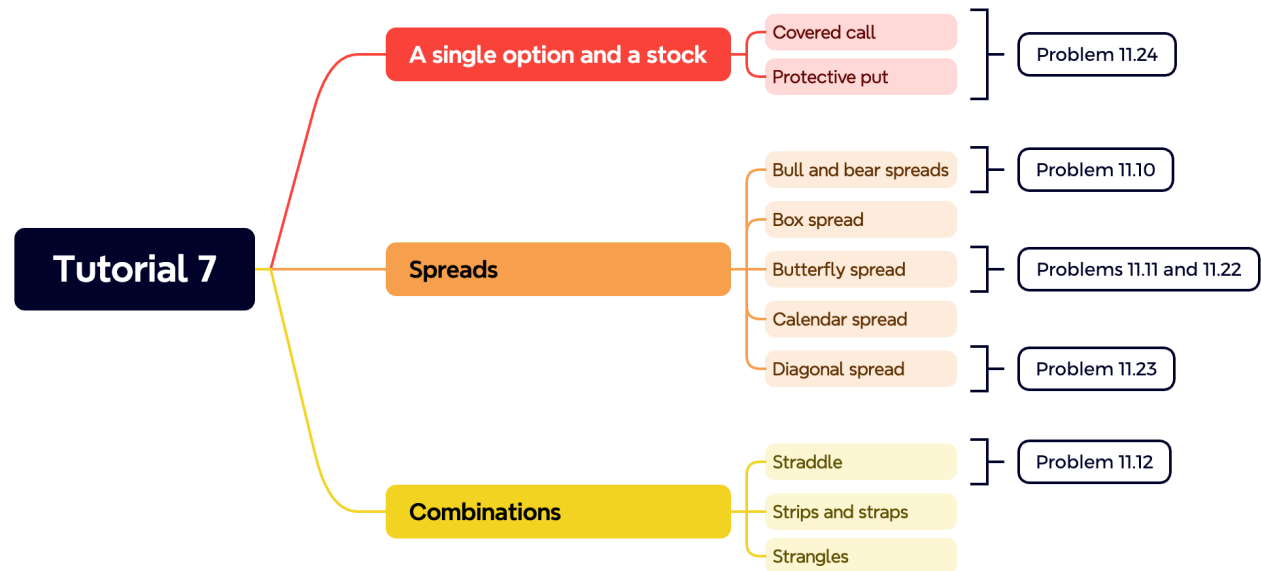
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Sep 2026

Agenda

- Problem 11.10
- Problem 11.11
- Problem 11.12
- Problem 11.22
- Problem 11.23
- Problem 11.24



Presented with xmind

Recap: Types of strategies

1. A single option and a stock
 - Covered call; protective put
2. Spreads (2+ calls **or** puts)
 - Bull and bear spreads; box spreads; butterfly spreads; calendar spreads; diagonal spreads
3. Combinations (2+ calls **and** puts)
 - Straddles; strips and straps; strangles

Build a desired payoff shape by adding simpler positions together!

Recap: Types of strategies

		View on underlying asset		
		Positive	Neutral	Negative
View on volatility	Positive	Buy call	Buy straddle Buy strangle Sell butterfly	Buy put
	Neutral	Buy asset Bull spread	Buy calendar spread	Sell asset Bear spread
	Negative	Sell put	Sell straddle Sell strangle Buy butterfly	Sell call

 NVIDIA Corp
NASDAQ: NVDA

How would you play Nvidia next?

222.27 USD ↑ 555,575.00% +222.23 all time

Closed: 18 Sept, 8:01pm GMT-4 • [Disclaimer](#)

After hours 222.53 +0.26 (0.12%)

1D | 5D | 1M | 6M | YTD | 1Y | 5Y | Max

"I strongly believe Nvidia will skyrocket soon, and I want leveraged upside for this massive rally."

"I am convinced the AI hype is over and Nvidia's stock is about to crash violently."



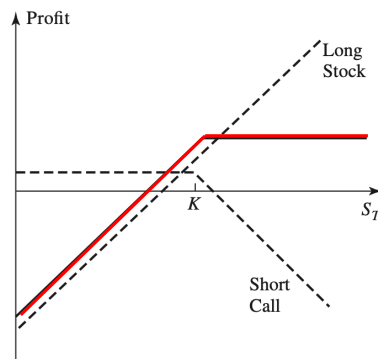
Open	219.35	Mkt cap	5.36T	Dividend	0.45%
High	222.73	P/E ratio	28.10	Qtrly div amt	0.25
Low	218.03	52-wk high	236.54	52-wk low	164.27

		View on Nvidia		
		Positive		Negative
View on volatility	High	<i>"I strongly believe Nvidia will skyrocket soon, and I want leveraged upside for this massive rally."</i> Buy call	<i>"Nvidia's upcoming earnings will trigger a massive shockwave, and I don't care if it crashes or moons as long as it explodes."</i> Buy straddle	<i>"I am convinced the AI hype is over and Nvidia's stock is about to crash violently."</i> Buy put
		<i>"Nvidia is a solid company that will steadily grind higher without any wild parabolic spikes".</i> Bull spread	<i>"Nvidia is just going to chop around in a boring, narrow range while the market digests recent news."</i> Buy calendar spread	<i>"Nvidia's momentum has cooled off, and I expect it to experience a moderate, controlled pullback."</i> Bear spread
	Low	<i>"Nvidia definitely won't crash, and I am happy to collect premium while waiting to buy it at a discount".</i> Sell put	<i>"Nvidia is completely out of gas and will flatline perfectly around its current price level."</i> Buy butterfly	<i>"Nvidia has hit its absolute ceiling and won't rally anymore, so I will collect rent from those hoping for another breakout."</i> Sell call

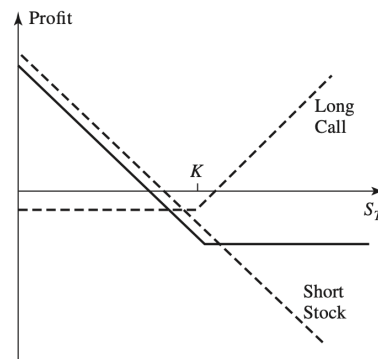
Recap: A single option and a stock

- Covered call (long a stock / short a call) → (a)
- Protective put (long a stock / long a put) → (c)

Synthetic short put!

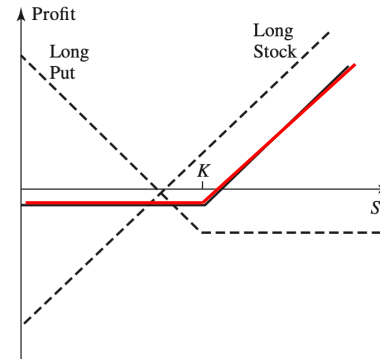


(a)

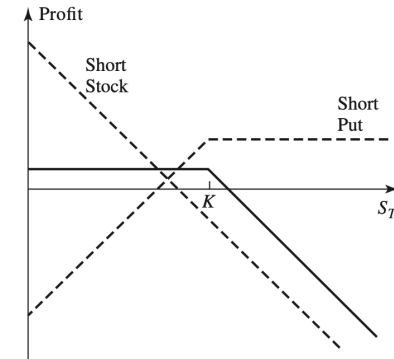


(b)

Synthetic long call!



(c)



(d)

Recap: A single option and a stock

The profit patterns in Figures 11.1a, b, c, and d have the same general shape as the profit patterns discussed in Chapter 9 for short put, long put, long call, and short call, respectively. Put-call parity provides a way of understanding why this is so. From Chapter 10, the put-call parity relationship is

Portfolio A: long put + long stock $\leftarrow$ $p + S_0 = c + Ke^{-rT} + D$ $\rightarrow$ Portfolio B: **long call** + cash (11.1)

protected put

Equation (11.1) shows that a long position in a European put combined with a long position in the stock is equivalent to a long European call position plus a certain amount ($= Ke^{-rT} + D$) of cash. This explains why the profit pattern in Figure 11.1c is similar to the profit pattern from a long call position. The position in Figure 11.1d is the reverse of that in Figure 11.1c and therefore leads to a profit pattern similar to that from a short call position.

Source: textbook, p270

Recap: A single option and a stock

Equation (11.1) can be rearranged to become

$$S_0 - c = Ke^{-rT} + D - p$$

This shows that a long position in a stock combined with a short position in a European call is equivalent to a short European put position plus a certain amount ($= Ke^{-rT} + D$) of cash. This explains why the profit pattern in Figure 11.1a is similar to that from a short put position. The position in Figure 11.1b is the reverse of that in Figure 11.1a and therefore leads to a profit pattern similar to that from a long put position.

Source: textbook, p271

Recap: A single option and a stock

Rules for function

Stocks

- Long stock
 - Payoff function = S_T
 - Profit & Loss (P&L) function = $S_T - S_0$
- Short sell stock
 - Payoff function = $-S_T$
 - Profit & Loss (P&L) function = $-S_T + S_0$

Payoff vs. profit (P&L): Payoff looks only at what happens at expiry. Profit also includes what we paid or received initially. Long positions pay the premium, while short positions receive the premium.

Recap: A single option and a stock

Options

Long call

- Payoff function = $\max(S - K, 0)$
- Profit & Loss (P&L) function = $\max(S - K, 0) - c$

• Short call

- Payoff function = $-\max(S - K, 0)$
- Profit & Loss (P&L) function = $-\max(S - K, 0) + c$

• Long put

- Payoff function = $\max(K - S, 0)$
- Profit & Loss (P&L) function = $\max(K - S, 0) - p$

• Short put

- Payoff function = $-\max(K - S, 0)$
- Profit & Loss (P&L) function = $-\max(K - S, 0) + p$

Recap: A single option and a stock

Rules for slope

- Slope tells us how much P&L changes when the stock price increases by \$1. Stock slope depends on the number of shares.

Stocks

- For 1 long share:

If S_T rises by \$1 → P&L rises by \$1; Slope=+1

- For n long shares:

If S_T rises by \$1 → P&L rises by \$ n ; Slope=+ n

Recap: A single option and a stock

Options

- Option slope depends on **both** the number of options and whether S_t is above or below the strike price K .

Position	$S_T < K$	$S_T \geq K$
n long calls	0	$+n$
n short calls	0	$-n$
n long puts	$-n$	0
n short puts	$+n$	0

More positions $\rightarrow$ Larger slope magnitude $\rightarrow$ Steeper P&L segment (where slope $\neq 0$)

Recap: A single option and a stock

Rule for initial cost ($S_0/\text{premium}$)

Position	Initial cash flow	P&L shift
n long shares	Pay nS_0	$-nS_0$
n short shares	Receive nS_0	$+nS_0$
n long calls	Pay nc	$-nc$
n short calls	Receive nc	$+nc$
n long puts	Pay np	$-np$
n short puts	Receive np	$+np$

- It shifts the P&L line **vertically** but does not change its slope.

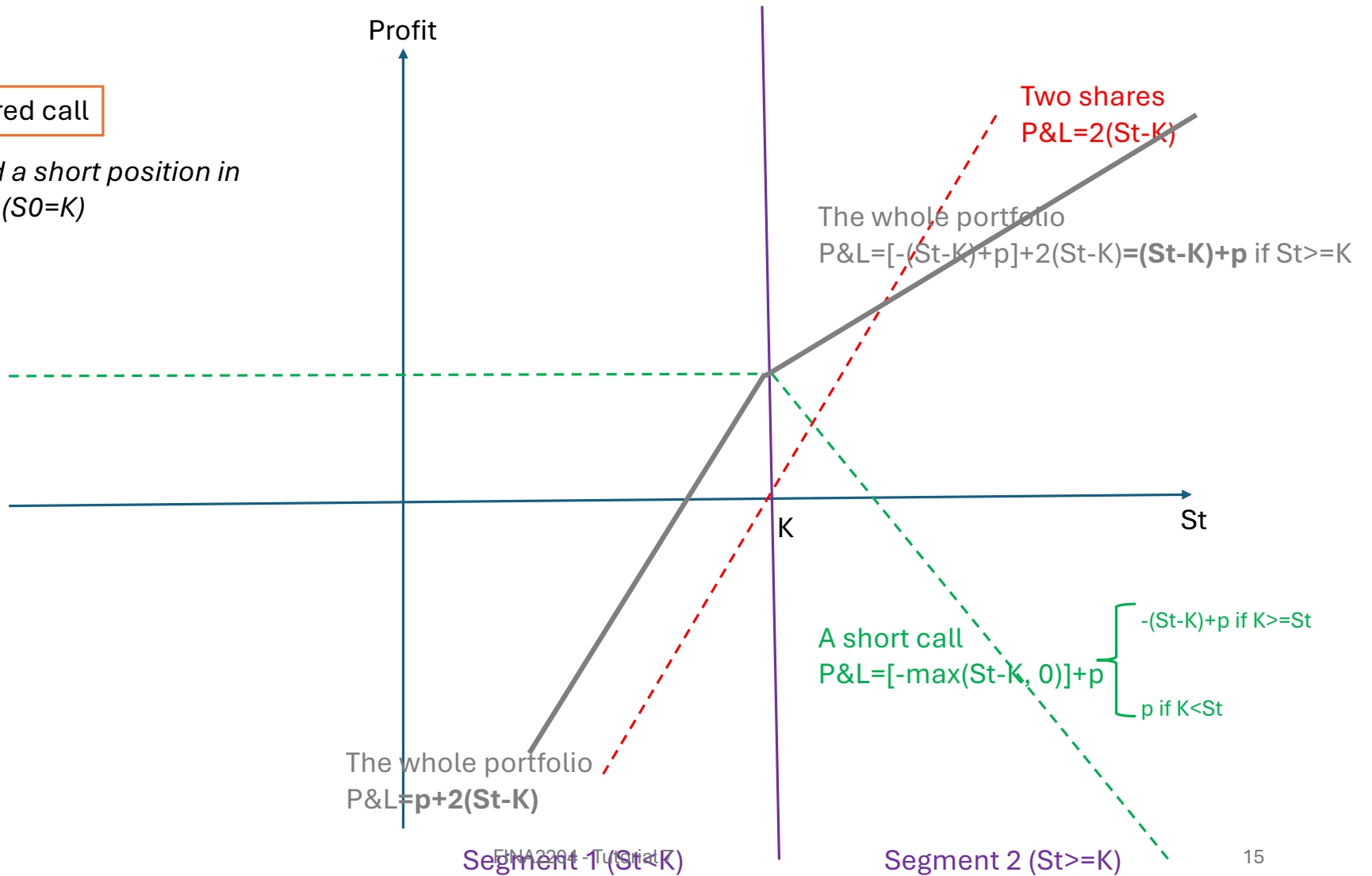
Recap: A single option and a stock

How to draw a portfolio diagram?

- 1. Plot individual components:** Draw the P&L line for each position separately using a *dashed line*, and write down their respective functions.
- 2. Define price segments:** Divide the x-axis into distinct segments based on the strike prices. Note that having n strike prices will always create $n+1$ segments.
- 3. Aggregate total profit:** Within each segment, sum the individual results and draw the total net profit as a *solid line*.

Covered call

Two shares and a short position in one call option ($S_0=K$)



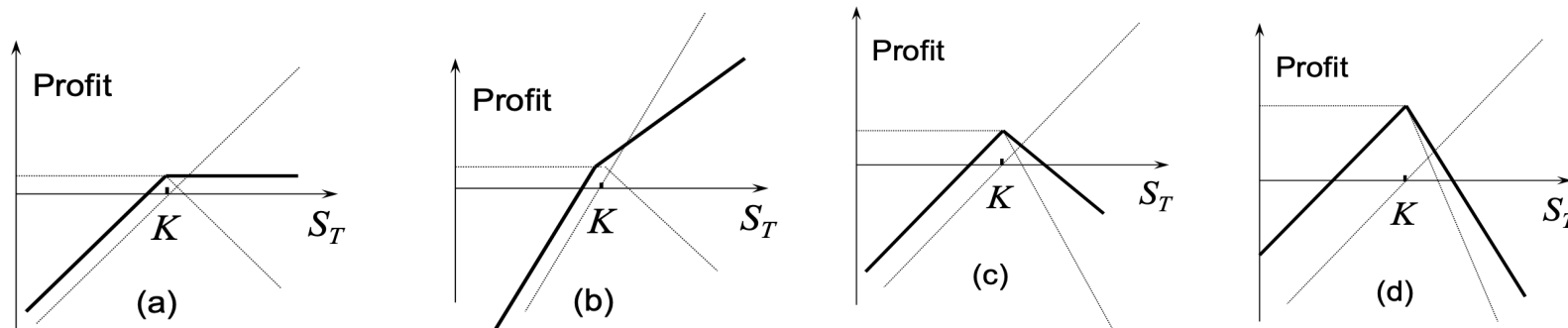
Stock price range	Payoff from long two stocks	Payoff from short call	Total Payoff	Profit
	(1)	(2)	(3)=(1)+(2)	(4)=(3)+ net upfront
$S_T < K$	$2S_T$	0	$2S_T$	$2S_T - 2K + p$
$S_T \geq K$	$2S_T$	$-(S_T - K)$	$2S_T - (S_T - K)$	$2S_T - (S_T - K) - 2K + p$

Problem 11.24

- *Draw a diagram showing the variation of an investor's profit and loss with the terminal stock price for a portfolio consisting of*
 - a. *One share and a short position in one call option*
 - b. *Two shares and a short position in one call option*
 - c. *One share and a short position in two call options*
 - d. *One share and a short position in four call options*
- *In each case, assume that the call option has an exercise price equal to the current stock price.*

Answer 11.24

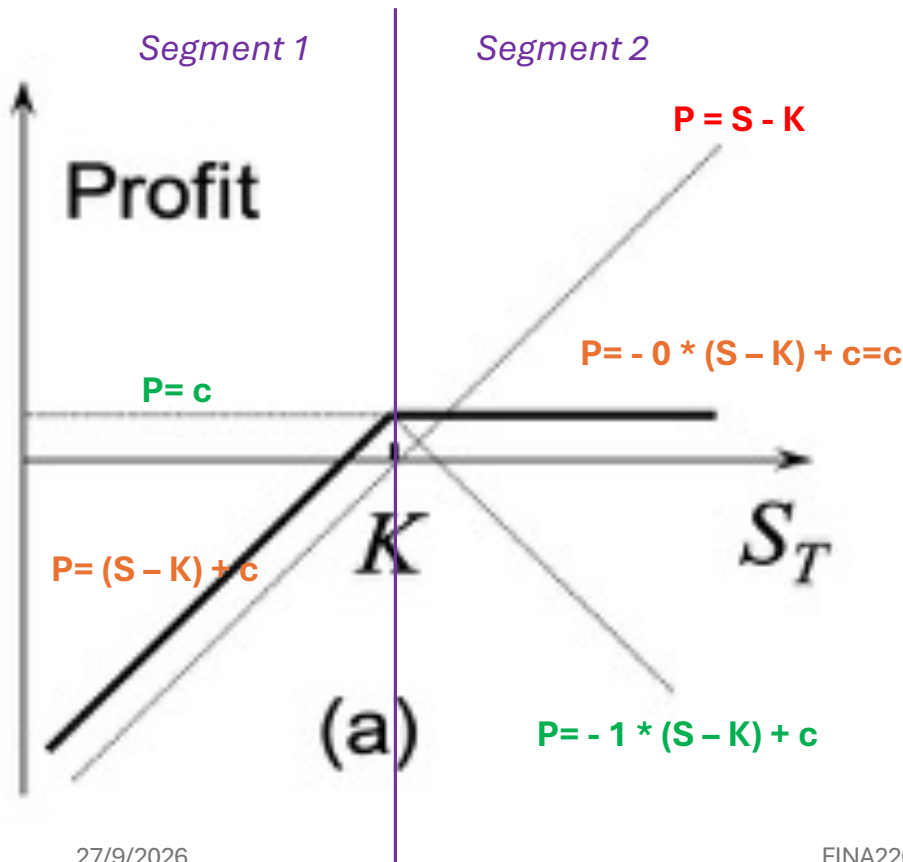
- The variation of an investor's profit/loss with the terminal stock price for each of the four strategies is shown in Figure below. In each case the dotted line shows the profits from the components of the investor's position and the solid line shows the total net profit.



Answer 11.24

Covered call

One share and a short position in **one** call options



The P&L function of one share:

$$S - K$$

The P&L function of selling one call option:

$$- [\max (S - K, 0)] + c$$

Or a short call

$$P = 0 + c, \text{ if } S < K$$

$$P = - (S - K) + c, \text{ if } S \geq K$$

The P&L function of total net profit:

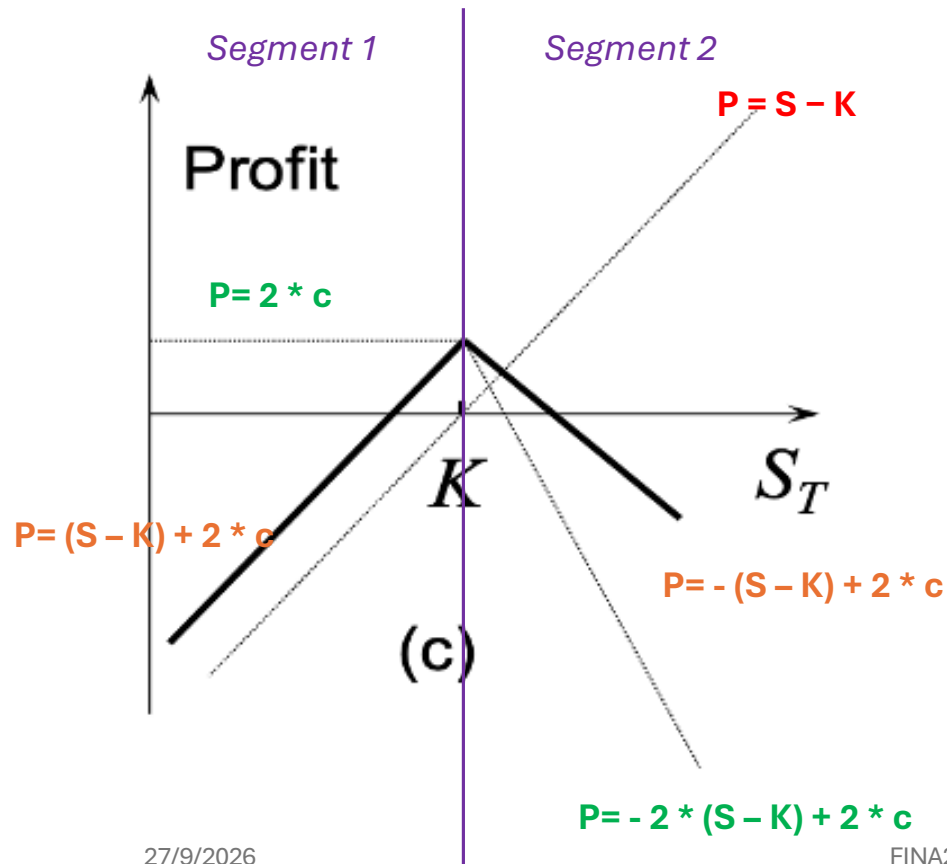
$$P = (S - K) + c, \text{ if } S < K$$

$$P = (S - K) + [- (S - K) + c], \text{ if } S \geq K$$

Answer 11.24

Covered call

One share and a short position in **two** call options



The P&L function of one share:
 $S - K$

The P&L function of selling four call options:
 $-2 * [\max(S - K, 0)] + 2 * c$

Or 2 short call

$$\begin{cases} P = 2 * c, \text{ if } S < K \\ P = -2 * (S - K) + 2 * c, \text{ if } S \geq K \end{cases}$$

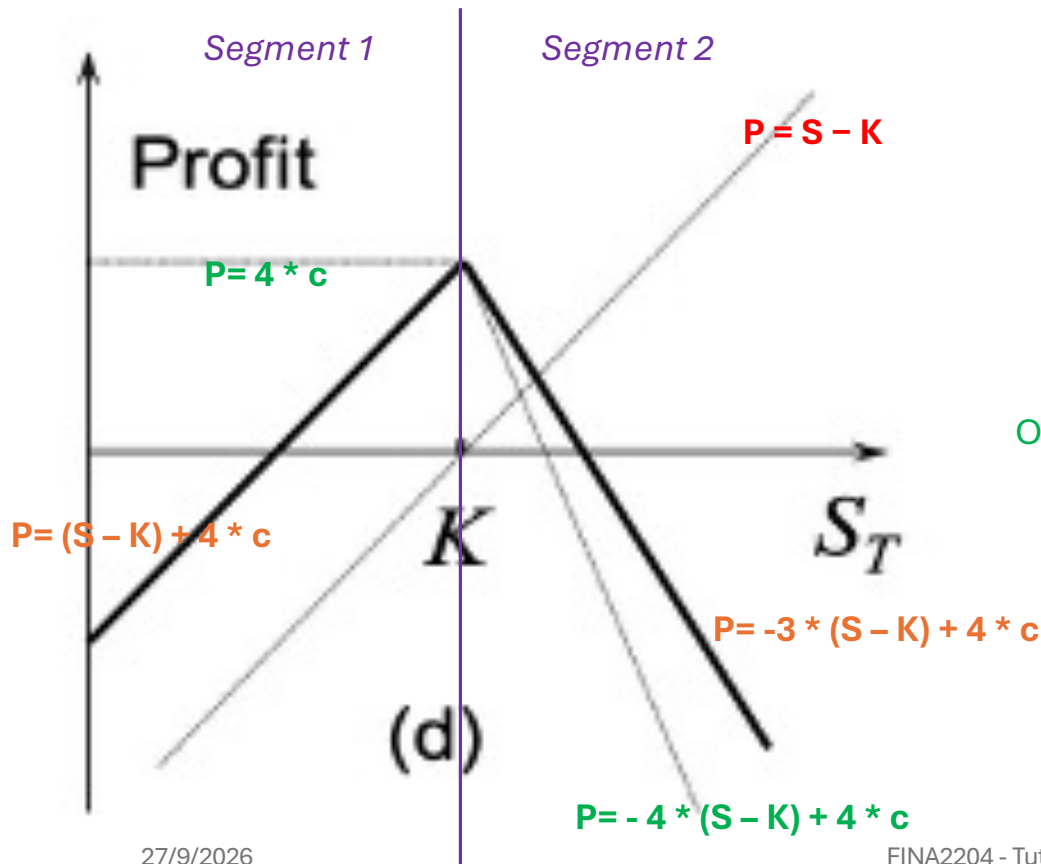
The P&L function of total net profit:

$$\begin{cases} P = (S - K) + 2 * c, \text{ if } S < K \\ P = (S - K) - 2 * (S - K) + 2 * c, \text{ if } S > K \end{cases}$$

Answer 11.24

Covered call

One share and a short position in **four** call options



The P&L function of one share:
 $S - K$

The P&L function of selling four call options:
 $-4 * [\max(S - K, 0)] + 4 * c$

Or 4 short call

$$\begin{cases} P = 4 * c, \text{ if } S < K \\ P = -4 * (S - K) + 4 * c, \text{ if } S \geq K \end{cases}$$

The P&L function of total net profit:

$$\begin{cases} P = (S - K) + 4 * c, \text{ if } S < K \\ P = (S - K) - 4 * (S - K) + 4 * c, \text{ if } S > K \end{cases}^{21}$$

Recap: Spreads

- A spread trading strategy involves taking a position in two or more options of **the same type** (i.e., two or more calls or two or more puts).
- One of the most popular types of spreads is a ***bull spread***. This can be created by buying a European call option on a stock with a certain strike price (K_1) and selling a European call option on the same stock with a higher strike price (K_2).
- An investor who enters into a bull spread is hoping that the stock price will increase (**positive movement; neutral volatility**).

Recap: Spreads

Bull spreads (call)

- Long the lower-strike call at K_1
- Short higher-strike call at K_2
- Net **cost** upfront (all else, equal, the value of a call option is negatively related to its strike price)

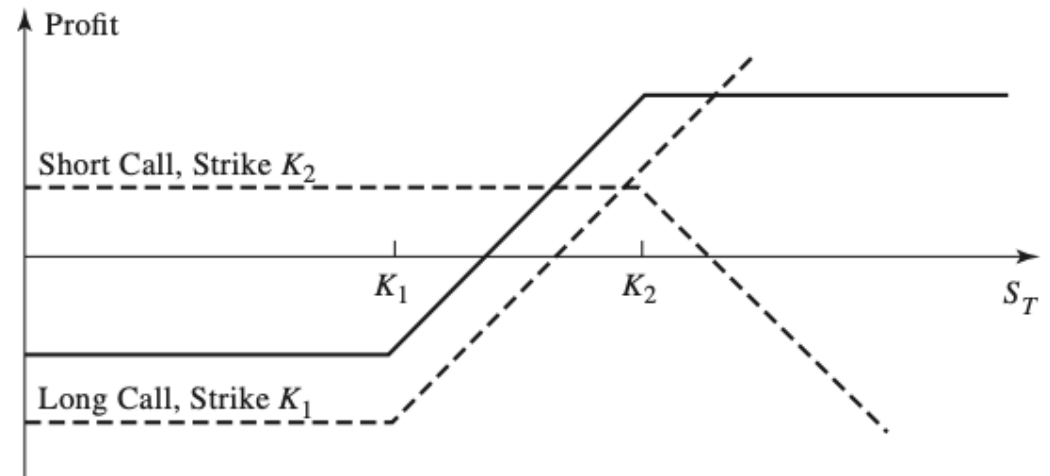


Figure 11.2 Profit from bull spread created using call options

Source: textbook, p271

Recap: Spreads

Bull spreads (put)

- Long put at K_1
- Short put at K_2
- Net **inflow** upfront (all else, equal, the value of a call option is positively related to its strike price)

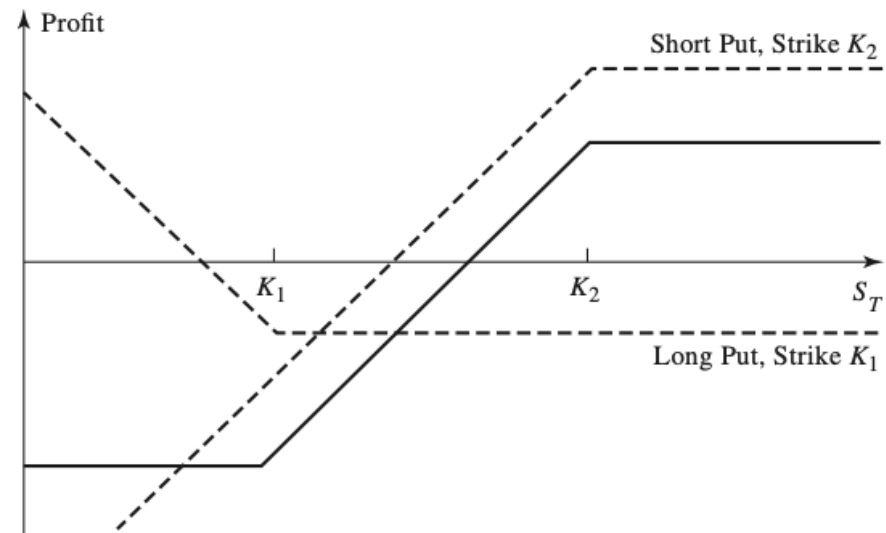


Figure 11.3 Profit from bull spread created using put options

Source: textbook, p273

Recap: Spreads

- By contrast, an investor who enters into a ***bear spread*** is hoping that the stock price will decline.
- The strike price of the option purchased is greater than the strike price of the option sold. (This is in contrast to a bull spread where the strike price of the option purchased is always less than the strike price of the option sold.)
- Market outlook: Both strategies **limit** the investor's upside as well as downside risk and suits an outlook of stability.

Recap: Spreads

Bear spreads (call)

- short the lower-strike call at K_1
- long the higher-strike call at K_2
- Net inflow upfront

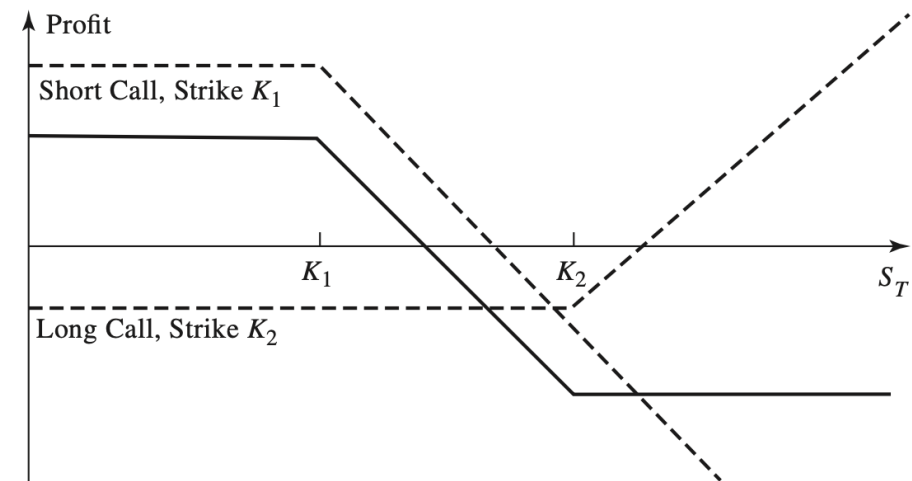


Figure 11.5 Profit from bear spread created using call options

Source: textbook, p274

Recap: Spreads

Bear spreads (put)

- short put at K_1
- long put at K_2
- Net cost upfront

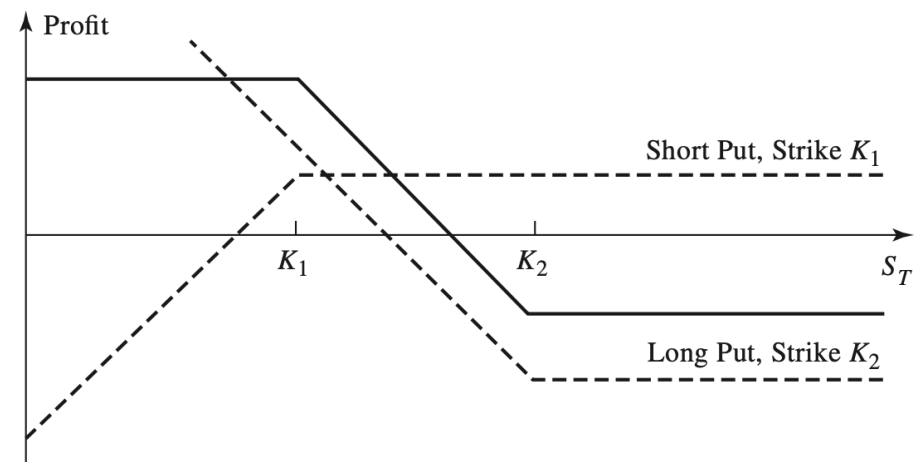


Figure 11.4 Profit from bear spread created using put options

Source: textbook, p273

Problem 11.10

- *Suppose that put options on a stock with strike prices \$30 and \$35 cost \$4 and \$7, respectively. How can the options be used to create (a) a bull spread and (b) a bear spread? Construct a table that shows the profit and payoff for both spreads.*

Answer 11.10

Setup

$K1 = \$30$

$K2 = \$35$

$p1 = \$4$; $p2 = \$7$

1. Bull spread (**buy** the \$30 put at \$4, **sell** the \$35 put at \$7)

> Net upfront = $7 - 4 = +\$3$

2. Bear spread (**sell** the \$30 put at \$4, **buy** the \$35 put at \$7)

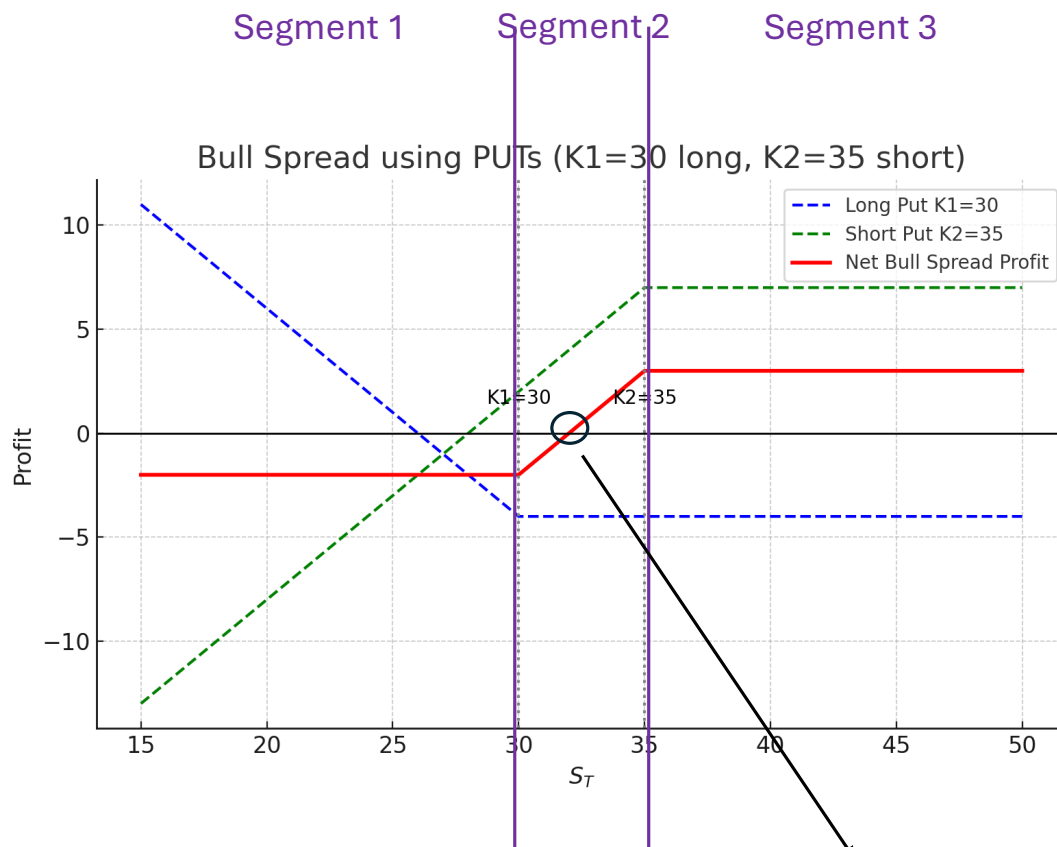
> Net upfront = $4 - 7 = -\$3$

Answer 11.10

- A bull spread (put) is created by buying the \$30 put and selling the \$35 put. This strategy gives rise to an **initial cash inflow** of \$3 (= \$7 - \$4). The outcome is as follows:

Stock price range	Payoff from long put	Payoff from short put	Total Payoff	Profit
	(1)	(2)	(3)=(1)+(2)	(4)=(3)+net upfront
$S_T < 30$	$30 - S_T$	$-(35 - S_T)$	-5	$-5 + 3$
$30 \leq S_T < 35$	0	$-(35 - S_T)$	$S_T - 35$	$S_T - 35 + 3$
$S_T \geq 35$	0	0	0	$0 + 3$

Answer 11.10



27/9/2026

FINA2204 - Tutorial 7
Break-even point = \$32

The P&L function of buying one put option:
 $\max(K1 - S, 0) - p1$

Or a long put

$$\begin{cases} P = (K1 - S) - p1, & \text{if } S < K1 \\ P = -p1, & \text{if } K1 \leq S < K2 \\ P = -p1, & \text{if } S \geq K2 \end{cases}$$

The P&L function of selling one put option:
 $- [\max(K2 - S, 0) - p2]$

Or a short put

$$\begin{cases} P = -[(K2 - S) - p2], & \text{if } S < K1 \\ P = -[(K2 - S) - p2], & \text{if } K1 \leq S < K2 \\ P = p2, & \text{if } S \geq K2 \end{cases}$$

The P&L function of total net profit:

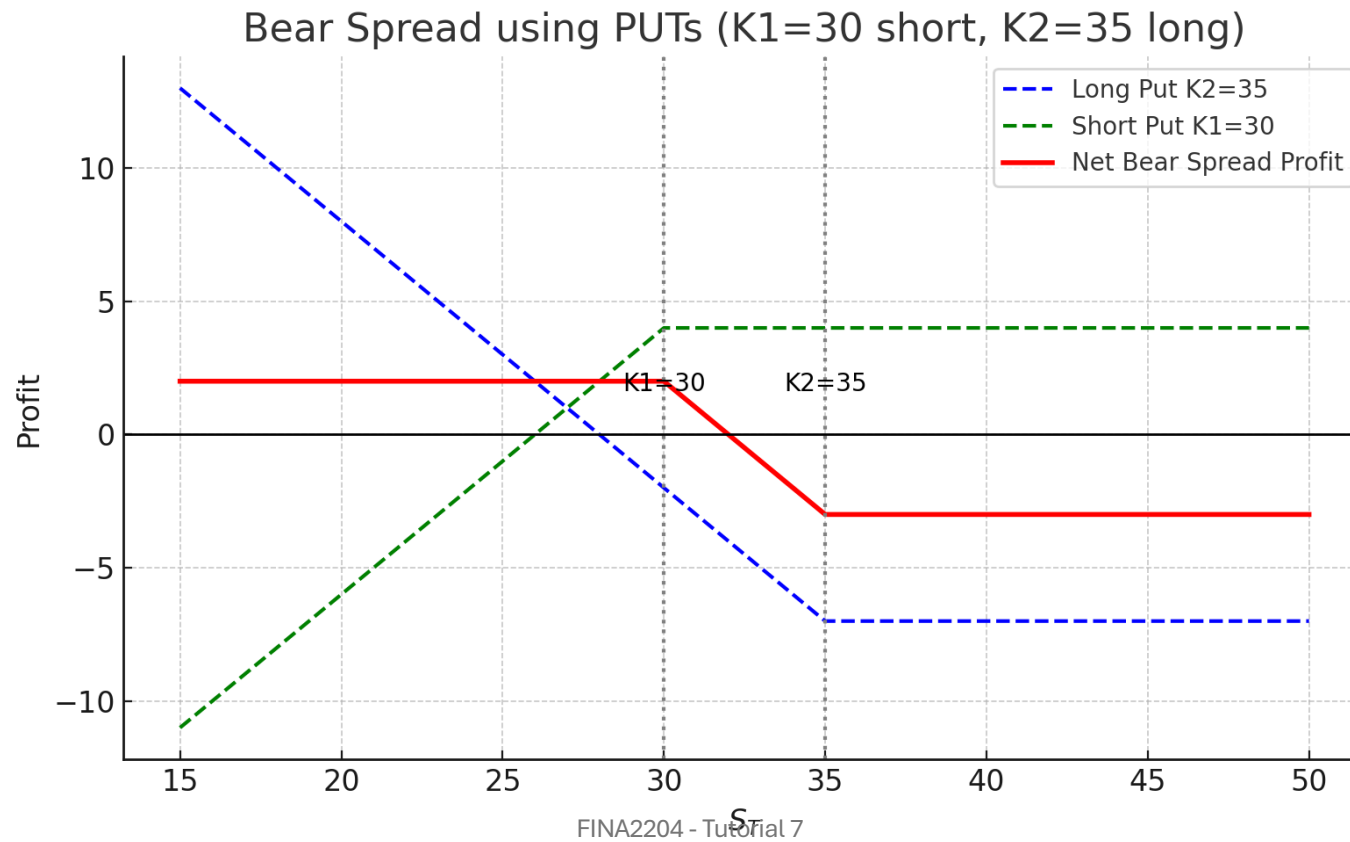
$$\begin{cases} P = (K1 - S) - p1 - [(K2 - S) - p2], & \text{if } S < K1 \\ P = -p1 - [(K2 - S) - p2], & \text{if } K1 \leq S < K2 \\ P = -p1 + p2, & \text{if } S \geq K2 \end{cases}$$

Answer 11.10

- A bear spread (put) is created by selling the \$30 put and buying the \$35 put. This strategy **costs** \$3 initially. The outcome is as follows:

Stock price range	Payoff from long put	Payoff from short put	Total Payoff	Profit
	(1)	(2)	(3)=(1)+(2)	(4)=(3)+net upfront
$S_T < 30$	$35 - S_T$	$-(30 - S_T)$	5	$5 - 3$
$30 \leq S_T < 35$	$35 - S_T$	0	$35 - S_T$	$(35 - S_T) - 3$
$S_T \geq 35$	0	0	0	$0 - 3$

Answer 11.10



Recap: Spreads

Box spreads

- A box spread is a combination of a bull call spread with strike prices K_1 and K_2 and a bear put spread with **the same two** strike prices.

Table 11.3 Payoff from a box spread

<i>Stock price range</i>	<i>Payoff from bull call spread</i>	<i>Payoff from bear put spread</i>	<i>Total payoff</i>
$S_T \leq K_1$	0	$K_2 - K_1$	$K_2 - K_1$
$K_1 < S_T < K_2$	$S_T - K_1$	$K_2 - S_T$	$K_2 - K_1$
$S_T \geq K_2$	$K_2 - K_1$	0	$K_2 - K_1$

The value of a box spread is therefore always the present value of this payoff or $(K_2 - K_1)e^{-rT}$. If it has a different value, there is an **arbitrage opportunity**.

Alpha Architect 1-3 Month Box ETF BOXX ★★ Morningstar Medalist Rating

Performance | Medalist Rating as of Jul 31, 2026 | [See Alpha Architect Investment Hub](#) >

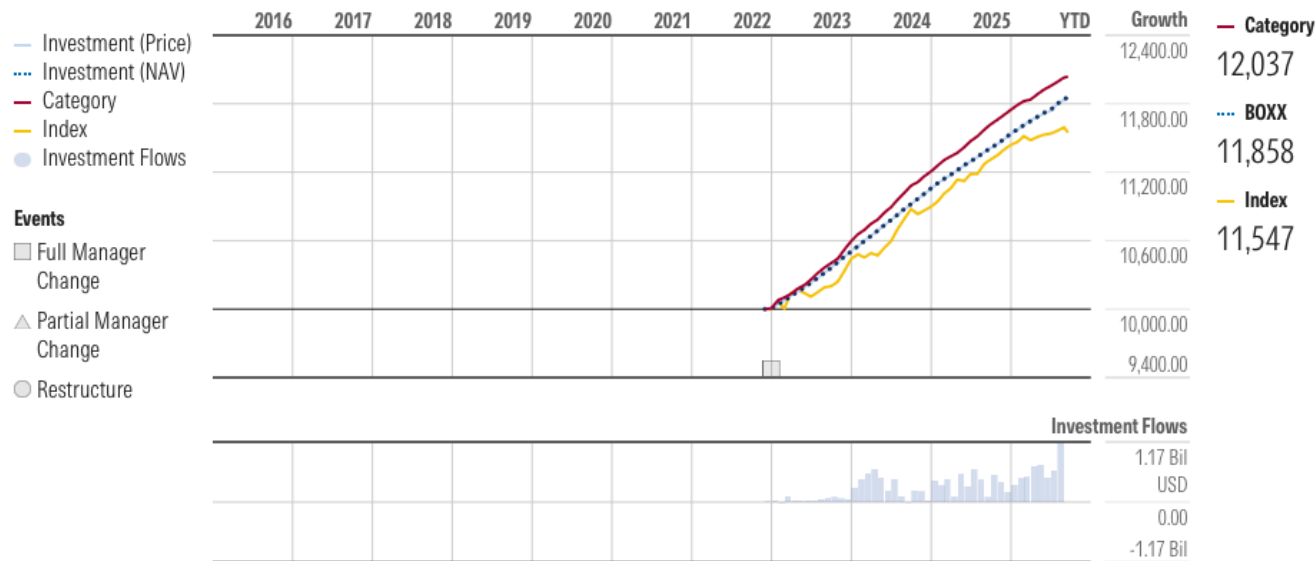


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Performance

Returns [Distributions](#)

Growth of 10,000



27/9/2026

Source: <https://www.morningstar.com/etfs/bats/boxx/quote>

35

Recap: Spreads

Butterfly spreads

- A butterfly spread involves positions in options with **three** different strike prices.
- It can be created by buying a European call option with a relatively low strike price, K_1 , buying a European call option with a relatively high strike price, K_3 , and selling two European call options with a strike price, K_2 , **halfway** between K_1 and K_3 . Generally K_2 is close to the current stock price.
- Butterfly spreads can also be created using put options with a similar strategy.

Recap: Spreads

- A butterfly spread leads to a profit if the stock price stays close to K_2 , but gives rise to a small loss if there is a significant stock price move in either direction (K_1, K_3).
- Market outlook: It is therefore an appropriate strategy for an investor who feels that large stock price moves are unlikely (**neutral movement; low volatility**).

Recap: Spreads

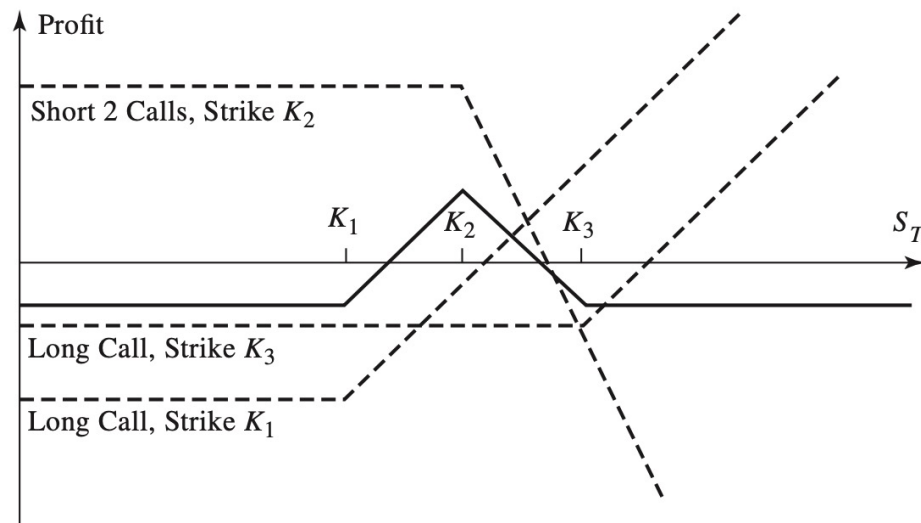


Figure 11.6 Profit from butterfly spread using call options

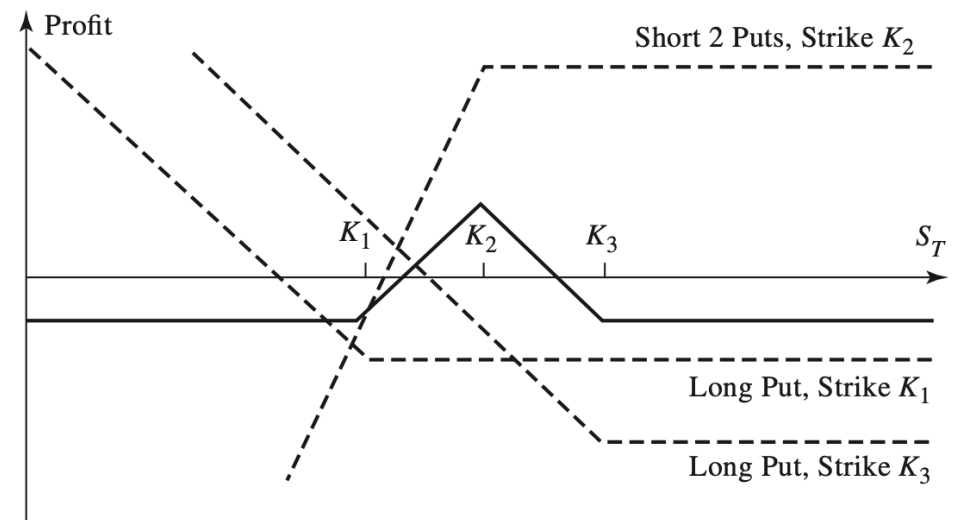


Figure 11.7 Profit from butterfly spread using put options

Source: textbook, p276-277

Recap: Spreads

- Unlike bull or bear spreads, which can start with either a cost or a cash inflow, a butterfly spread always requires **a net cost upfront** (The premium from selling two middle-strike options is never enough to cover the cost of the two wings, so the strategy is always set up as a debit position).
- Whether built with calls or puts, a butterfly spread always has **the same upfront cost**. By put–call parity, the option prices align so that call and put butterflies yield identical payoff profiles, ensuring their initial premiums are equal.

Problem 11.11

- *Use put–call parity to show that the cost of a butterfly spread created from European puts is identical to the cost of a butterfly spread created from European calls.*

Answer 11.11

- Define c_1, c_2 and c_3 as the prices of calls with strike prices K_1, K_2 and K_3 . Define p_1, p_2 and p_3 as the prices of puts with strike prices K_1, K_2 and K_3 . With the usual notation

$$c_1 + K_1 e^{-rT} = p_1 + S \quad (1)$$

Put-call parity

$$c_2 + K_2 e^{-rT} = p_2 + S \quad (2)$$

$$c_3 + K_3 e^{-rT} = p_3 + S \quad (3)$$

$$K_2 = \frac{(K_1 + K_3)}{2} \quad (4)$$

Hence, combining (1), (2), and (3):

$$(1) + (3) - 2 \times (2) \rightarrow c_1 + c_3 - 2c_2 + (K_1 + K_3 - 2K_2)e^{-rT} = p_1 + p_3 - 2p_2$$

Answer 11.11

Because (4), it follows that $K_1 + K_3 - 2K_2 = 0$ and

$$c_1 + c_3 - 2c_2 = p_1 + p_3 - 2p_2$$

- The cost of a butterfly spread created using European calls is therefore **exactly the same as** the cost of a butterfly spread created using European puts.

Problem 11.22

- *Three put options on a stock have the same expiration date and strike prices of \$55, \$60, and \$65. The market prices are \$3, \$5, and \$8, respectively. Explain how a butterfly spread can be created. Construct a table showing the profit from the strategy. For what range of stock prices would the butterfly spread lead to a loss?*

Answer 11.22

- A butterfly spread is created by buying the \$55 put, buying the \$65 put and selling two of the \$60 puts. This costs $3 + 8 - 2 \times 5 = \$1$ initially. The following table shows the profit/loss from the strategy.

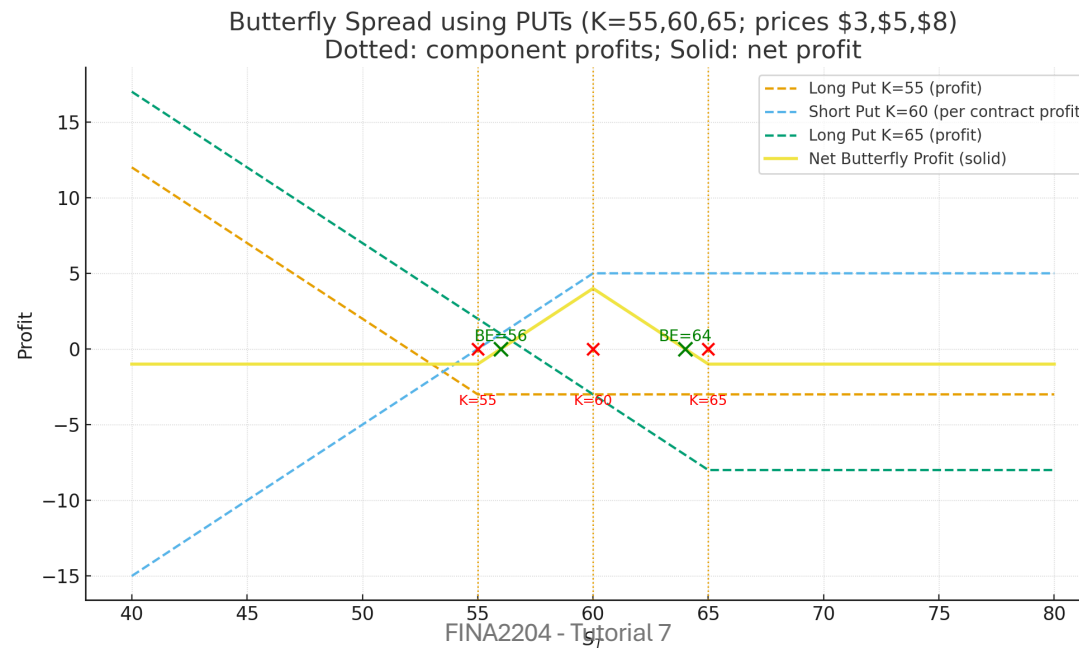
Answer 11.22

Stock price range	Payoff from long put	Payoff from short put	Total Payoff	Profit
$S_T < 55$	$(55 - S_T) + (65 - S_T)$	$-2 \times (60 - S_T)$	0	$0 - 1$
$55 \leq S_T < 60$	$65 - S_T$	$-2 \times (60 - S_T)$	$S_T - 55$	$(S_T - 55) - 1$
$60 \leq S_T < 65$	$65 - S_T$	0	0	$(65 - S_T) - 1$
$S_T \geq 65$	0	0	0	$0 - 1$

$$\left\{ \begin{array}{l} (S - 55) - 1 < 0, S < 56 \\ (65 - S) - 1 < 0, S > 64 \end{array} \right. \longrightarrow \mathbf{S < 56, \text{ and } 64 < S}$$

Answer 11.22

- The butterfly spread leads to a loss when the final stock price is greater than \$64 or less than \$56.



Recap: Spreads

Calendar spreads

- The options have the same strike price and different expiration dates ($T_1 < T_2$). The pattern is **similar to** the profit from the butterfly spread.

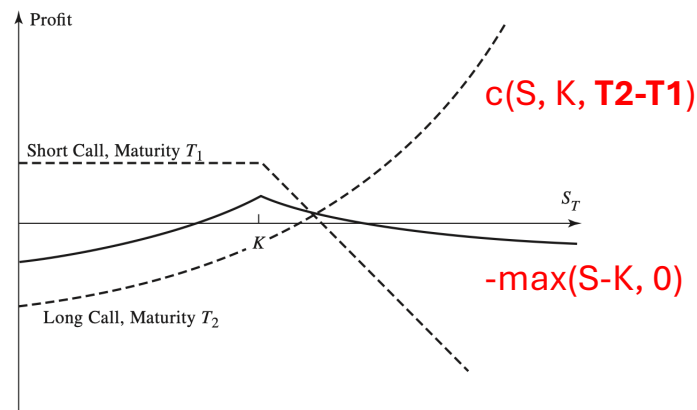


Figure 11.8 Profit from calendar spread created using two calls when $T_2 > T_1$, calculated at the time when the short maturity call expires

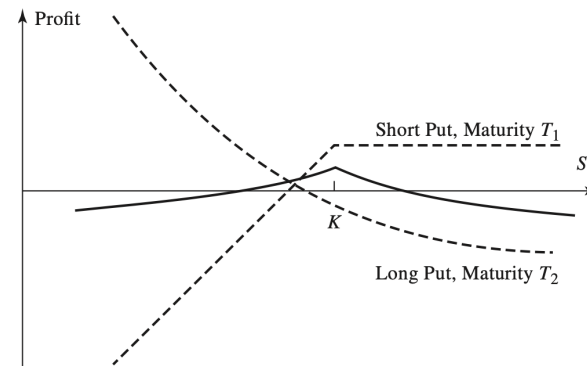


Figure 11.9 Profit from calendar spread created using two puts when $T_2 > T_1$, calculated at the time when the short maturity put expires

Source: textbook, p278

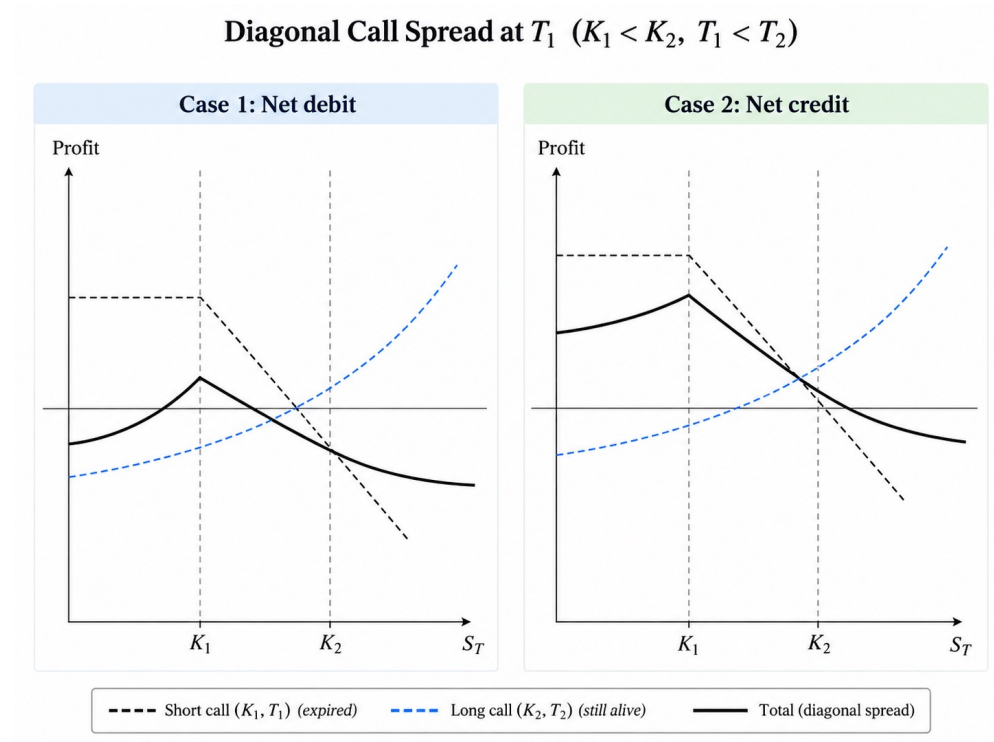
Recap: Spreads

Diagonal spreads

- A diagonal spread involves two options of the same type with **different** strike prices and **different** expiration dates ($K_1 \neq K_2, T_1 < T_2$)
- Strategies: *short call* (K_1, T_1) + *long call* (K_2, T_2)
 1. Longer maturity ($T_1 < T_2$) → tend to make the long call more valuable.
 2. Higher strike price ($K_1 < K_2$) → tend to make the long call less valuable.

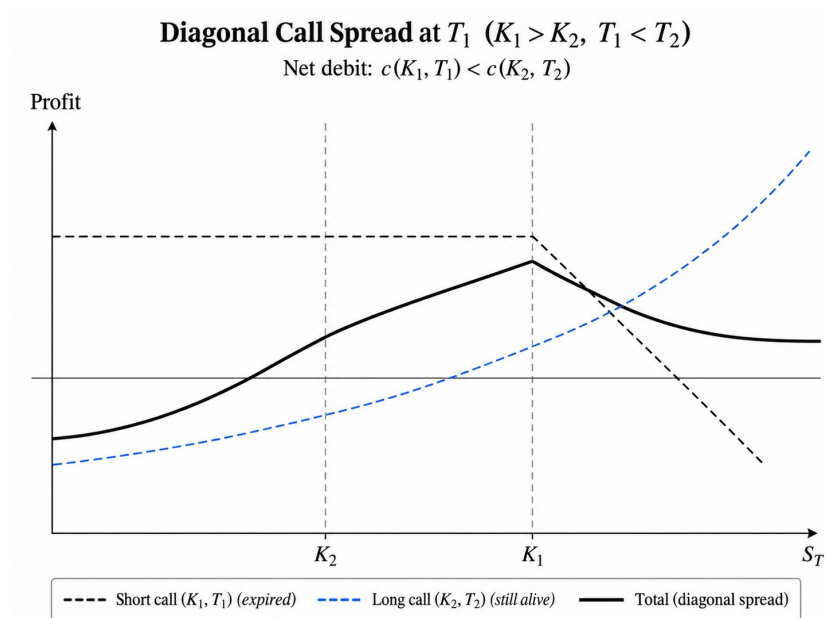
Recap: Spreads

- $K_1 < K_2, T_1 < T_2 \rightarrow c(K_1, T_1) ? c(K_2, T_2)$
- Which option has the higher premium is **ambiguous!**
- *Case 1:* $c(K_1, T_1) < c(K_2, T_2)$
- *Case 2:* $c(K_1, T_1) > c(K_2, T_2)$



Recap: Spreads

- $K_1 > K_2, T_1 < T_2 \rightarrow c(K_1, T_1) < c(K_2, T_2)$



Problem 11.23

- *A diagonal spread is created by buying a call with strike price K_2 and exercise date T_2 and selling a call with strike price K_1 and exercise date T_1 ($T_2 > T_1$). Draw a diagram showing the the value of the spread at time T_1 when (a) $K_2 > K_1$, and (b) $K_2 < K_1$.*

Answer 11.23

- There are two alternative profit patterns for part (a). These are shown in Figures S11.2 and S11.3. In Figure S11.2 the long maturity (high strike price) option is worth more than the short maturity (low strike price) option. In Figure S11.3 the reverse is true. There is no ambiguity about the profit pattern for part (b). This is shown in Figure S11.4.

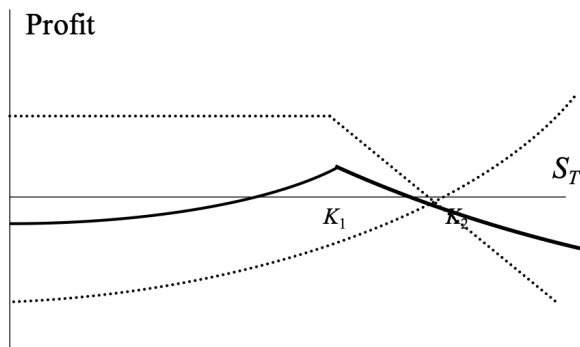


Figure S11.2: Investor's Profit/Loss in Problem 11.20a when long maturity call is worth more than short maturity call

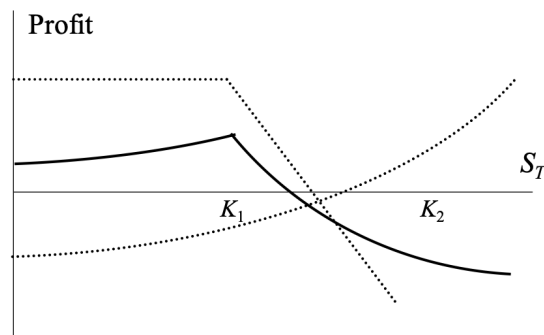


Figure S11.3 Investor's Profit/Loss in Problem 11.20b when short maturity call is worth more than long maturity call

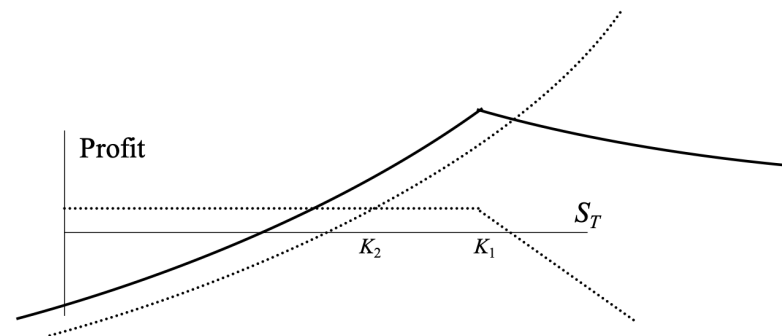


Figure S11.4 Investor's Profit/Loss in Problem 11.20b

Recap: Combinations

- A combination is an option trading strategy that involves taking a position in **both** calls and puts on the same stock. We will consider straddles, strips, straps, and strangles.

Straddle

- One popular combination is a straddle, which involves buying one European call and **one** European put with the same strike price and expiration date.
- Because we are buying both options, we pay both premiums upfront (**net cost upfront**).

Recap: Combinations

- Market belief: A straddle is appropriate when an investor is expecting a large move in a stock price but does not know in which direction the move will be (**neutral movement; high volatility**).

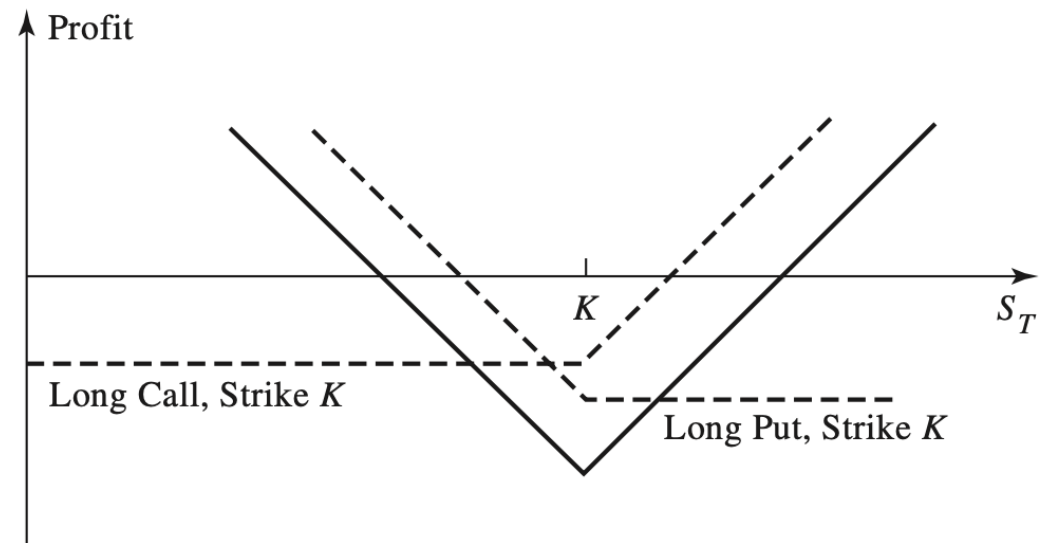


Figure 11.10 Profit from a straddle

Source: textbook, p279

Recap: Combinations

Strips and Straps

- A strip consists of a long position in one European call and **two** European puts with the same strike price and expiration date.
- A strip is similar to a straddle, but it is **asymmetric**.
- Market belief: In a strip the investor is betting that there will be a big stock price move and considers a decrease in the stock price to be **more likely than** an increase.

Recap: Combinations

- A strap simply reverses that asymmetry.
- A strap consists of a long position in **two** European calls and **one** European put with the same strike price and expiration date.
- Market belief: In a strap the investor is also betting that there will be a big stock price move. However, in this case, an increase in the stock price is considered to be **more likely than** a decrease.

Recap: Combinations

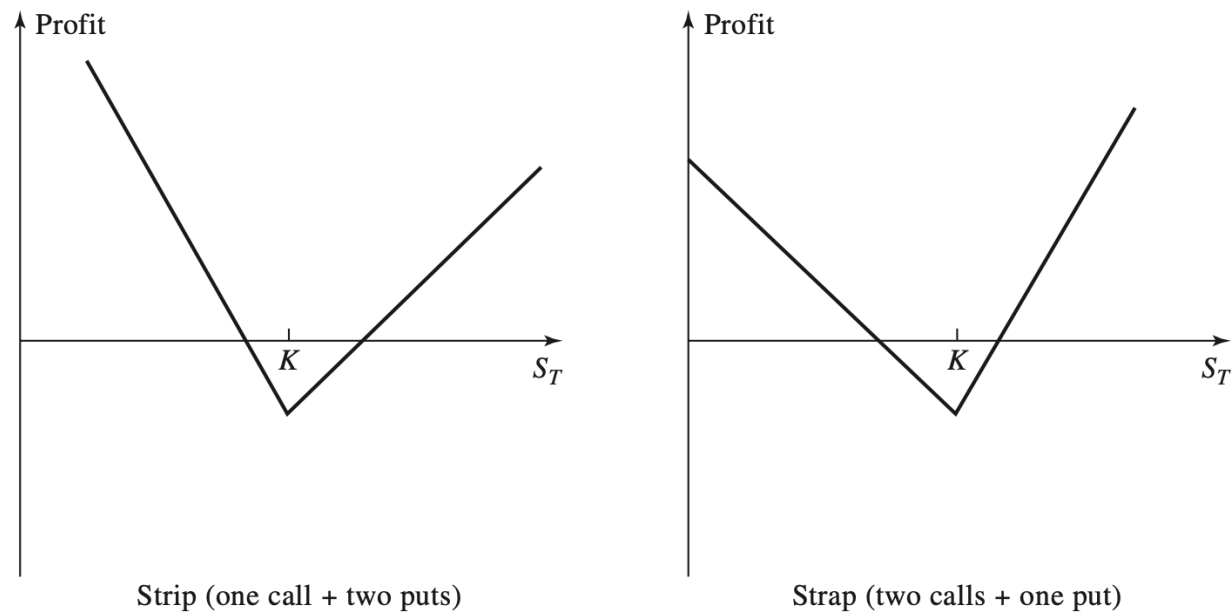


Figure 11.11 Profit from a strip and a strap

Source: textbook, p280

Recap: Combinations

Strangles

- Definition: In a strangle, sometimes called a *bottom vertical combination*, an investor buys a European put and a European call with the same expiration date and **different** strike prices.
- Market belief: A strangle is a similar strategy to a straddle. The investor is betting that there will be **a large price move**, but is **uncertain** whether it will be an increase or a decrease.
- The sale of a strangle is suitable when an investor expects little price movement. Like selling a straddle, it is risky because potential losses are unlimited.

Recap: Combinations

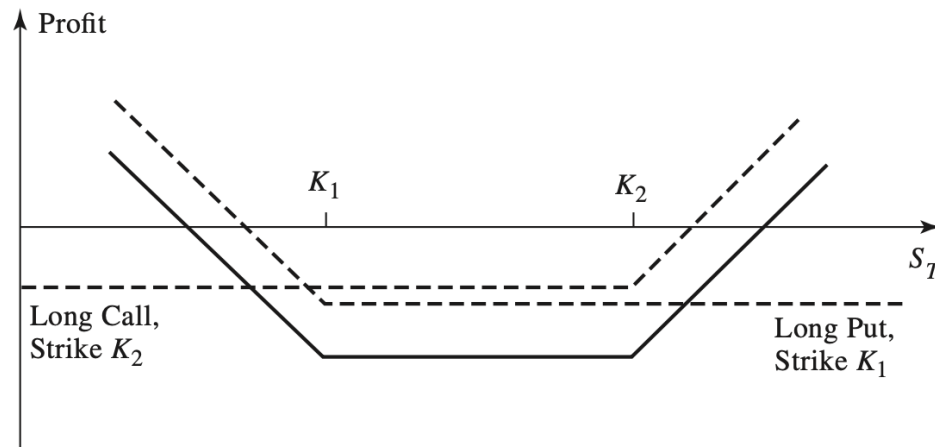


Table 11.6 Payoff from a strangle

<i>Range of stock price</i>	<i>Payoff from call</i>	<i>Payoff from put</i>	<i>Total payoff</i>
$S_T \leq K_1$	0	$K_1 - S_T$	$K_1 - S_T$
$K_1 < S_T < K_2$	0	0	0
$S_T \geq K_2$	$S_T - K_2$	0	$S_T - K_2$

Source: textbook, p281-282

Problem 11.12

- *A call with a strike price of \$60 costs \$6. A put with the same strike price and expiration date costs \$4. Construct a table that shows the profit from a straddle. For what range of stock prices would the straddle lead to a loss?*

Answer 11.12

- A straddle is created by buying both the call and the put. This strategy costs \$10. The profit/loss is shown in the following table:

Stock price range	Payoff from long call	Payoff from long put	Total Payoff	Profit
$S_T \leq 60$	0	$60 - S_T$	$60 - S_T$	$(60 - S_T) - 10$
$S_T > 60$	$S_T - 60$	0	$S_T - 60$	$(S_T - 60) - 10$

$$\left\{ \begin{array}{l} (60 - S) - 10 < 0, S > 50 \\ (S - 60) - 10 < 0, S < 70 \end{array} \right. \longrightarrow 50 < S < 70$$

Answer 11.12

- This shows that the straddle will lead to a loss if the final stock price is between \$50 and \$70.

